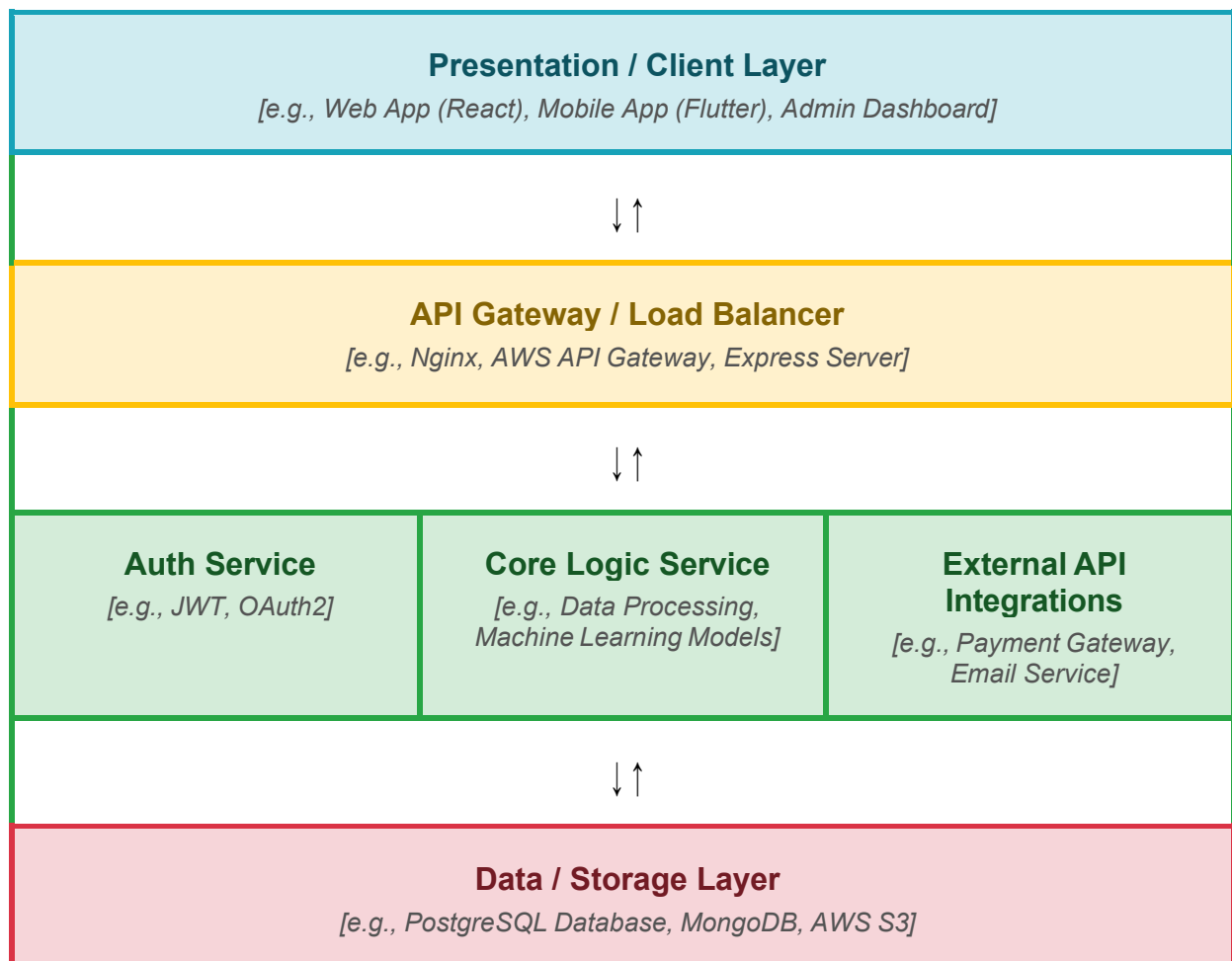


Solution Architecture

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user interface for entering weather parameters and displaying flood prediction results.	HTML, CSS, Bootstrap, JavaScript
API Gateway	Receives user requests and connects the web interface with the prediction services.	Flask
Microservices	Processes data and executes machine learning models to predict flood occurrence.	Python, Scikit-learn, XGBoost
Database	Stores historical datasets, trained models, and prediction data.	CSV Files, Pandas DataFrames